

## SAFETY DATA SHEET

Creation Date 29-October-2010

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** Isobutyric acid

**Cat No. :** AC122520000; AC122520010; AC122520250; AC122525000

**CAS-No** 79-31-2  
**Synonyms** 2-Methylpropionic acid

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Flammable liquids</b>                                | Category 3   |
| <b>Acute oral toxicity</b>                              | Category 4   |
| <b>Acute dermal toxicity</b>                            | Category 3   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Flammable liquid and vapor  
Harmful if swallowed  
Toxic in contact with skin  
Causes severe skin burns and eye damage  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting/equipment  
Use only non-sparking tools  
Take precautionary measures against static discharges  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Stench

### 3. Composition/Information on Ingredients

| Component       | CAS-No  | Weight % |
|-----------------|---------|----------|
| Isobutyric acid | 79-31-2 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical

|                                        |                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | advice.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                          |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.                                                       |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                       |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 59 °C / 138.2 °F                                                                                                                        |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 420 °C / 788 °F                                                                                                                         |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 9.2%                                                                                                                                    |
| <b>Lower</b>                            | 2%                                                                                                                                      |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b> | Should not be released into the environment. Do not flush into surface water or sanitary sewer system.                                                                                                                                               |

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Corrosives area. Incompatible Materials. Bases. Strong oxidizing agents. Reducing Agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

### Engineering Measures

Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) Goggles

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                            |
|----------------------------------------|--------------------------------------------|
| Physical State                         | Liquid                                     |
| Appearance                             | Colorless                                  |
| Odor                                   | Stench                                     |
| Odor Threshold                         | No information available                   |
| pH                                     | 2.9 (0.1M)                                 |
| Melting Point/Range                    | -47 °C / -52.6 °F                          |
| Boiling Point/Range                    | 153 - 154 °C / 307.4 - 309.2 °F @ 760 mmHg |
| Flash Point                            | 59 °C / 138.2 °F                           |
| Evaporation Rate                       | No information available                   |
| Flammability (solid,gas)               | Not applicable                             |
| Flammability or explosive limits       |                                            |
| Upper                                  | 9.2%                                       |
| Lower                                  | 2%                                         |
| Vapor Pressure                         | 0.43 mmHg @ 20 °C                          |
| Vapor Density                          | 3.04 (Air = 1.0)                           |
| Specific Gravity                       | 0.950                                      |
| Solubility                             | No information available                   |
| Partition coefficient; n-octanol/water | No data available                          |
| Autoignition Temperature               | 420 °C / 788 °F                            |
| Decomposition Temperature              | No information available                   |
| Viscosity                              | 1.1 mPa s at 20 °C                         |
| Molecular Formula                      | C4 H8 O2                                   |
| Molecular Weight                       | 88.11                                      |

## 10. Stability and reactivity

|                                  |                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                            |
| Stability                        | Stable under normal conditions.                                                                       |
| Conditions to Avoid              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Bases, Strong oxidizing agents, Reducing Agent                                                        |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                               |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                              |
| Hazardous Reactions              | None under normal processing.                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component       | LD50 Oral                               | LD50 Dermal          | LC50 Inhalation         |
|-----------------|-----------------------------------------|----------------------|-------------------------|
| Isobutyric acid | 266 mg/kg ( Rat )<br>2230 mg/kg ( Rat ) | 475 mg/kg ( Rabbit ) | > 0.55 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component       | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------|---------|------------|------------|------------|------------|------------|
| Isobutyric acid | 79-31-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Algae                                  | Freshwater Fish | Microtox            | Water Flea                                     |
|-----------------|---------------------------------------------------|-----------------|---------------------|------------------------------------------------|
| Isobutyric acid | EC50: = 45 mg/L, 72h<br>(Desmodesmus subspicatus) | Not listed      | EC50 = 57 mg/L 17 h | EC50: = 51 mg/L, 48h<br>(Daphnia magna Straus) |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component       | log Pow |
|-----------------|---------|
| Isobutyric acid | 0.88    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2529  
 Proper Shipping Name ISOBUTYRIC ACID  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group III

### TDG

UN-No UN2529  
 Proper Shipping Name ISOBUTYRIC ACID

|                                |                 |
|--------------------------------|-----------------|
| <b>Hazard Class</b>            | 3               |
| <b>Subsidiary Hazard Class</b> | 8               |
| <b>Packing Group</b>           | III             |
| <b>IATA</b>                    |                 |
| <b>UN-No</b>                   | UN2529          |
| <b>Proper Shipping Name</b>    | ISOBUTYRIC ACID |
| <b>Hazard Class</b>            | 3               |
| <b>Subsidiary Hazard Class</b> | 8               |
| <b>Packing Group</b>           | III             |
| <b>IMDG/IMO</b>                |                 |
| <b>UN-No</b>                   | UN2529          |
| <b>Proper Shipping Name</b>    | ISOBUTYRIC ACID |
| <b>Hazard Class</b>            | 3               |
| <b>Subsidiary Hazard Class</b> | 8               |
| <b>Packing Group</b>           | III             |

## 15. Regulatory information

### International Inventories

| Component       | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Isobutyric acid | 79-31-2 | X   | -    | X    | ACTIVE                                        | 201-195-7 | -      | -   |

| Component       | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------|---------|-------|----------|------|------|------|------|-------|-------|
| Isobutyric acid | 79-31-2 | X     | KE-24875 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component       | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Isobutyric acid | Part 4 Substance                                     |                                                                            |                                           |

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component       | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Isobutyric acid | 79-31-2 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component       | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Isobutyric acid | 79-31-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y34                      |

## 16. Other information

|                  |                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared By      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                 |
| Creation Date    | 29-October-2010                                                                                                                                                                    |
| Revision Date    | 24-December-2021                                                                                                                                                                   |
| Print Date       | 24-December-2021                                                                                                                                                                   |
| Revision Summary | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**